PYTHON PROGRAMMING

CASE STUDY ASSIGNMENT – 5

Case Study 5 Attendance Recovery Computation.

1. Problem Statement

Students receive marks in different subjects during assessments. Comparing these marks manually can make it difficult to identify improvement or decline in performance. This project develops a Python program that compares a student's marks in two assessments, calculates the change in marks for each subject, identifies the subject with the lowest marks in the second assessment, and summarizes the student's overall performance trend.

2. Objectives

1. To record a student's marks in two assessments for multiple subjects.
2. To calculate the difference between the marks obtained in both assessments for each subject.
3. To identify subjects in which the student has improved or declined.
4. To identify the subject with the lowest marks in the second assessment.
5. To compare the total marks of both assessments and display the overall performance trend.

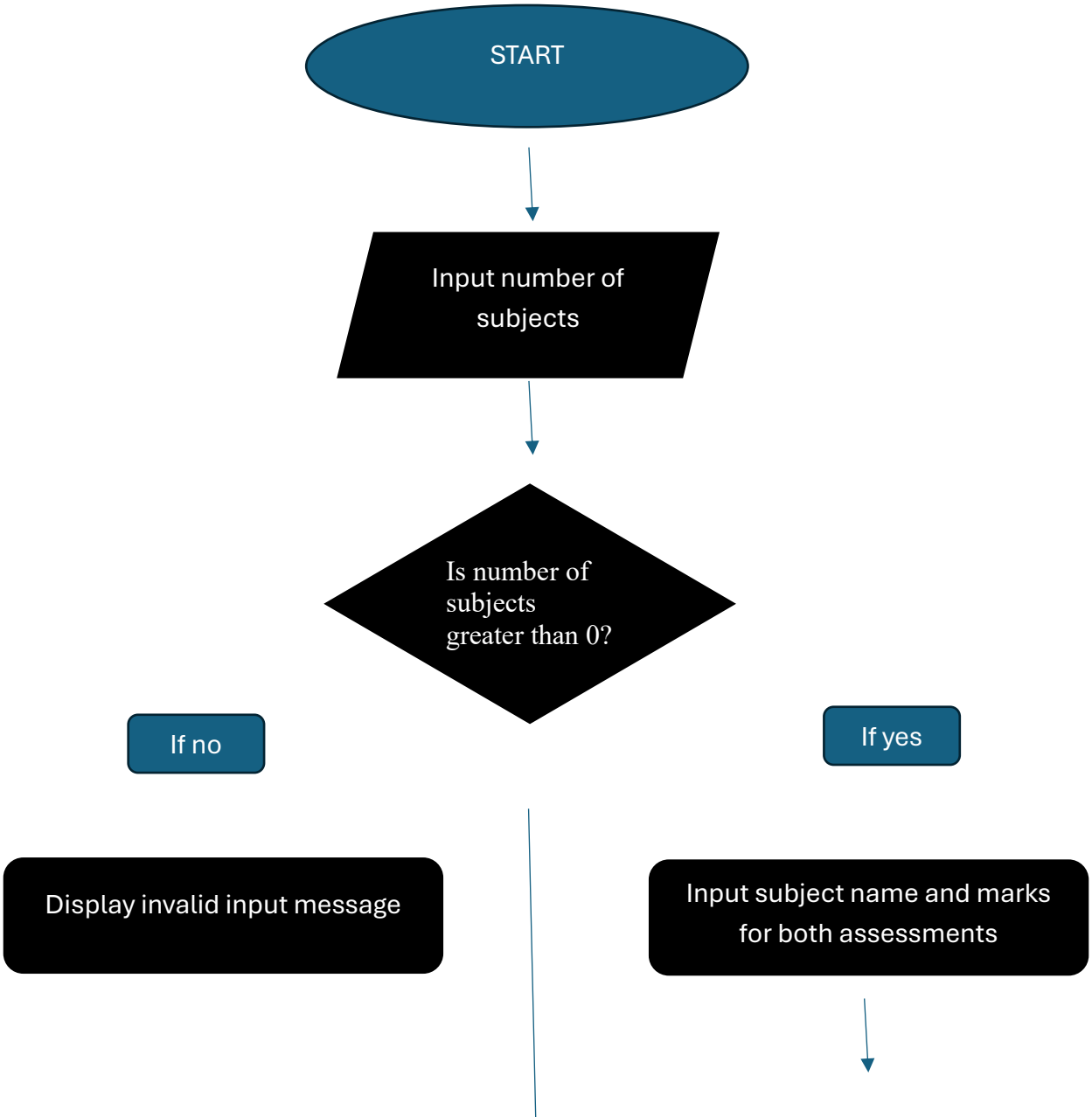
3. Assumptions

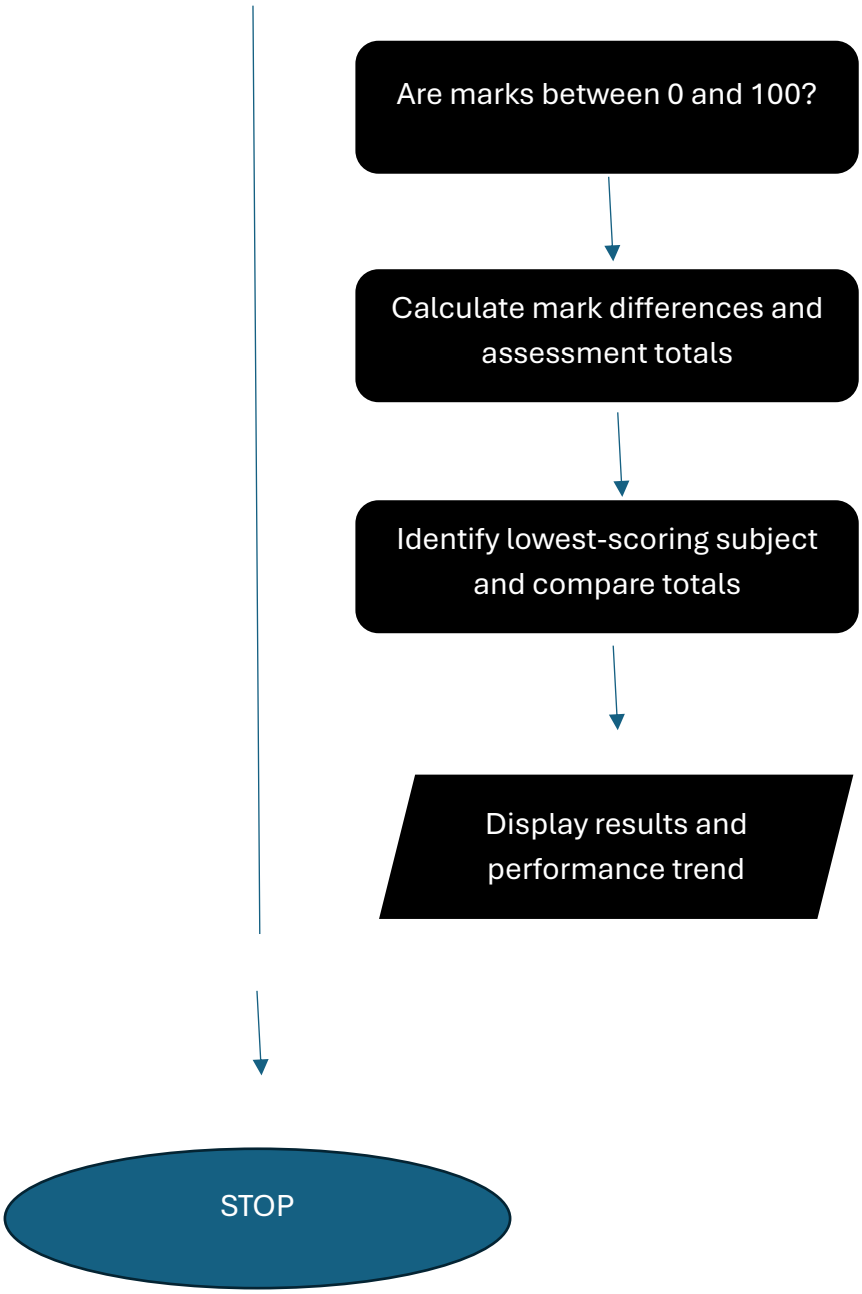
1. At least one subject must be entered.
2. Marks for each assessment must be between 0 and 100.
3. The same subjects are considered in both assessments.
4. The difference in marks is calculated by subtracting Assessment 1 marks from Assessment 2 marks.
5. If the total marks in Assessment 2 are greater than Assessment 1, performance is considered improved.
6. If both assessment totals are equal, performance is considered unchanged.

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5. Flowchart





6.pseudo code

START

INPUT number _of_ subjects

IF number_ of_ subjects <= 0 THEN

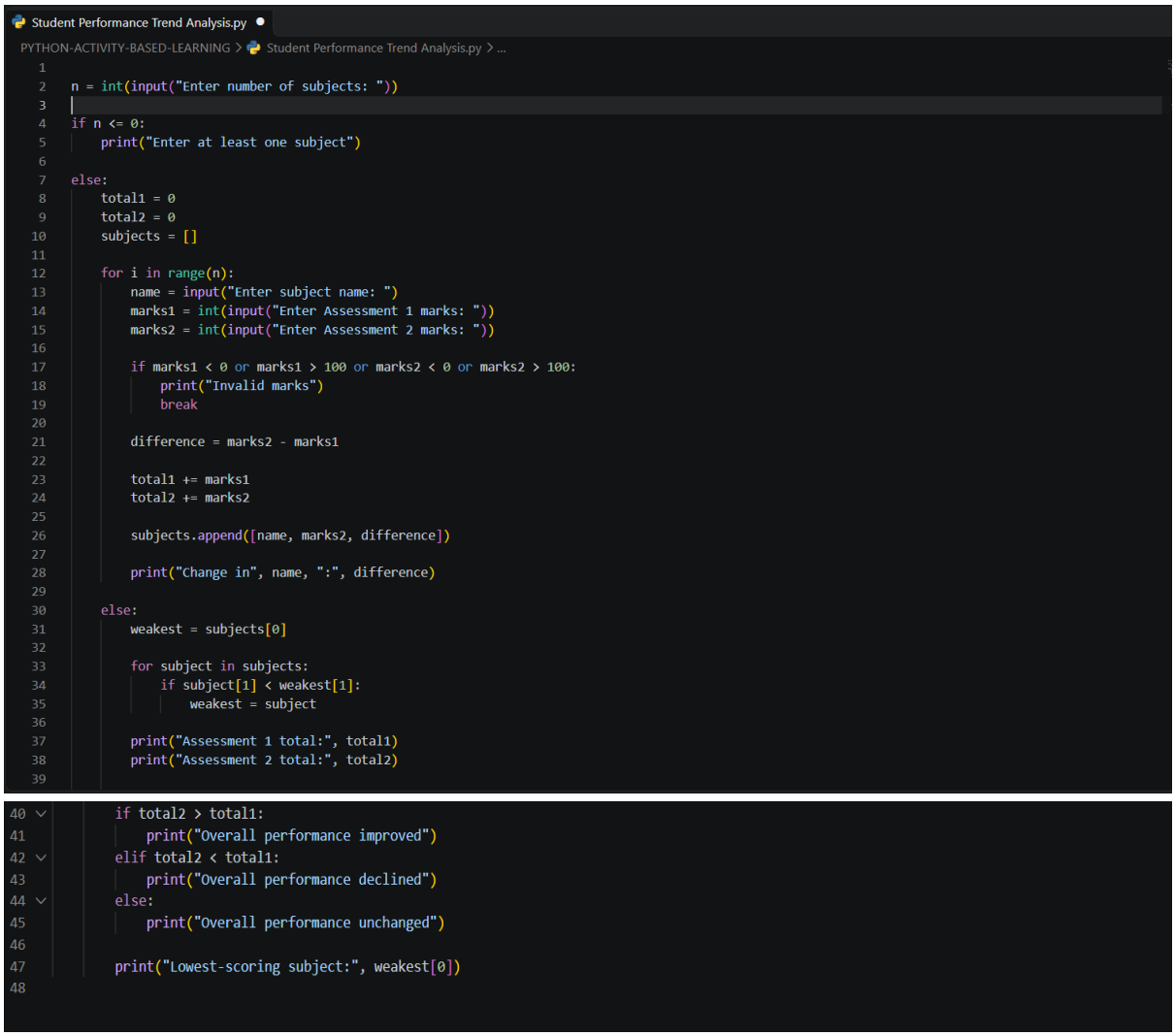
DISPLAY "Enter at least one subject"

ELSE

total1 = 0

```
total2 = 0
FOR each subject
  INPUT subject_ name
  INPUT marks1
  INPUT marks2
  IF marks1 < 0 OR marks1 > 100 OR marks2 < 0 OR marks2 > 100 THEN
    DISPLAY "Invalid marks"
    STOP
  END IF
  difference = marks2 - marks1
  total1 = total1 + marks1
  total2 = total2 + marks2
  DISPLAY subject_ name and difference
END FOR
IDENTIFY subject with lowest marks in Assessment 2
DISPLAY total1 and total2
IF total2 > total1 THEN
  DISPLAY "Overall performance improved"
ELSE IF total2 < total1 THEN
  DISPLAY "Overall performance declined"
ELSE
  DISPLAY "Overall performance unchanged"
END IF
```

7.source code



```
Student Performance Trend Analysis.py
PYTHON-ACTIVITY-BASED-LEARNING > Student Performance Trend Analysis.py > ...
n = int(input("Enter number of subjects: "))
if n <= 0:
    print("Enter at least one subject")

else:
    total1 = 0
    total2 = 0
    subjects = []

    for i in range(n):
        name = input("Enter subject name: ")
        marks1 = int(input("Enter Assessment 1 marks: "))
        marks2 = int(input("Enter Assessment 2 marks: "))

        if marks1 < 0 or marks1 > 100 or marks2 < 0 or marks2 > 100:
            print("Invalid marks")
            break

        difference = marks2 - marks1

        total1 += marks1
        total2 += marks2

        subjects.append([name, marks2, difference])

        print("Change in", name, ":", difference)

    else:
        weakest = subjects[0]

        for subject in subjects:
            if subject[1] < weakest[1]:
                weakest = subject

        print("Assessment 1 total:", total1)
        print("Assessment 2 total:", total2)

        if total2 > total1:
            print("Overall performance improved")
        elif total2 < total1:
            print("Overall performance declined")
        else:
            print("Overall performance unchanged")

        print("Lowest-scoring subject:", weakest[0])
```

8. Test Cases

Test Case	Input	Expected Output
1	Maths: 60 → 75, English: 70 → 65	Maths change: +15, English change: -5, Assessment 1 total: 130, Assessment 2 total: 140, Overall performance improved, Lowest-scoring subject: English

2	Maths: 80 → 70	Assessment 1 total: 80, Assessment 2 total: 70, Overall performance declined
3	Science: 75 → 75	Assessment 1 total: 75, Assessment 2 total: 75, Overall performance unchanged
4	Number of subjects: 0	Enter at least one subject
5	Maths marks: 110	Invalid marks

9. Output Screenshots

The output screenshots of the Student Performance Trend Analysis program will be attached below.

Screenshot 1: Program output showing improved performance.

```
PS D:\CODING> & C:\Users\ASUS\AppData\Local\Python\bin\python.exe "d:/CODING/PYTHON-ACTIVITY-BASED-LEARNING/Student Performance Trend Analysis.py
Enter number of subjects: 2
Enter subject name: maths
Enter Assessment 1 marks: 60
Enter Assessment 2 marks: 75
Change in maths : 15
Enter subject name: english
Enter Assessment 1 marks: 70
Enter Assessment 2 marks: 65
Change in english : -5
Assessment 1 total: 130
Assessment 2 total: 140
Overall performance improved
Lowest-scoring subject: english
```

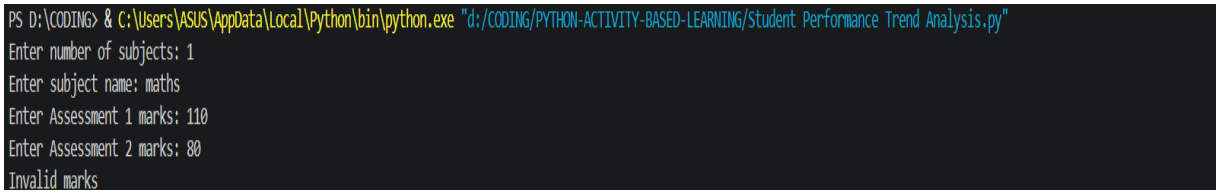
Screenshot 2: Program output showing declined performance.

```
PS D:\CODING> & C:\Users\ASUS\AppData\Local\Python\bin\python.exe "d:/CODING/PYTHON-ACTIVITY-BASED-LEARNING/Student Performance Trend Analysis.py"
Enter number of subjects: 1
Enter subject name: maths
Enter Assessment 1 marks: 80
Enter Assessment 2 marks: 70
Change in maths : -10
Assessment 1 total: 80
Assessment 2 total: 70
Overall performance declined
Lowest-scoring subject: maths
```

Screenshot 3: Program output showing unchanged performance.

```
PS D:\CODING> & C:\Users\ASUS\AppData\Local\Python\bin\python.exe "d:/CODING/PYTHON-ACTIVITY-BASED-LEARNING/Student Performance Trend Analysis.py"
Enter number of subjects: 1
Enter subject name: science
Enter Assessment 1 marks: 75
Enter Assessment 2 marks: 75
Change in science : 0
Assessment 1 total: 75
Assessment 2 total: 75
Overall performance unchanged
Lowest-scoring subject: science
```

Screenshot 4: Program output showing invalid marks.



10. Conclusion

The Student Performance Trend Analysis program was developed successfully using Python. It accepts students' marks in different subjects for two assessments, calculates the change in marks, compares the overall performance, and identifies the lowest-scoring subject. The program also validates marks to prevent invalid values. This project demonstrates the practical use of loops, conditional statements, lists, arithmetic operations, and input validation in Python. It helps analyze academic performance and identify subjects that may need additional attention.